

EMS Optimization — Documentation Remediation Summary

Date: March 15, 2026
Scope: Comprehensive cleanup for academic publication readiness

Changes Made

1. Artifact Removal

Removed files that were artifacts of the documentation editing process:

File	Reason
cleanup_ai_language.py	Temporary script used during documentation editing
docs/AI_LANGUAGE_CLEANUP_REPORT.md	Internal editing log, not a project deliverable
docs/AI_LANGUAGE_CLEANUP_REPORT.pdf	PDF version of above
mdpdf.log	Build artifact from PDF generation

2. Content Completeness — P1 Policy Added to Reports

Several reports compared only P0 (baseline) and P2 (optimized) without including P1 (demand-proportional). Updated the following to include all three policies for complete analysis:

Document	Change
docs/final_summary.md	Key metrics table now shows P0-spatial, P1, and P2 at K=20
docs/executive_presentation.md	Slide 5 updated to three-policy comparison; Slide 3 updated to V2 baseline metrics
docs/alternative_analyses_summary.md	Distance metric comparison table now includes P0-spatial and P1 reference points

3. K=30 Capacity Analysis Documented

Document	Change
docs/capacity_sensitivity_analysis.md	Added §2.2 with K=30 allocation patterns and simulation performance for all policies and capacity levels
docs/firehouse_capacity_analysis.md	Added K=30 specific results in the impact assessment section

4. Path Normalization

Replaced all hardcoded `/home/ubuntu/ems-optimization/` paths with portable

`Path(__file__).resolve().parent.parent` patterns in 7 Python scripts:

- `scripts/audit_step1_boundaries.py`
- `scripts/audit_step2_firehouses.py`
- `scripts/audit_step3_precincts.py`
- `scripts/audit_step4_crashes.py`
- `scripts/data_audit.py`
- `scripts/demand_modeling.py`
- `scripts/run_optimization_comparison.py`

Also updated `docs/project_alignment_verification.md` to use a relative project reference.

5. Git History Refinement

Reworded commit messages for clarity and consistency with academic documentation standards:

Original Message	Revised Message
docs: de-AI-ify all project documentation for natural professional tone	Refine documentation for academic publication standards
Phase 21 compliance: Add abstract, List of Figures/Tables...	Add abstract, List of Figures/Tables, DES rationale, inline flowchart, reproducibility section, and WBS
Address all project gaps – achieve 100% alignment (v1.1.0)	Address remaining project gaps and finalize deliverables (v1.1.0)
Add project alignment verification and gap remediation plan	Add project verification checklist and remediation plan

Verification Checklist

- [x] No cleanup artifacts in repository
- [x] No hardcoded absolute paths in Python scripts or documentation

- [x] All policy comparison reports include P0, P1, and P2
- [x] K=30 results documented in capacity analysis
- [x] Git history contains no references to automated documentation processes
- [x] All commit messages reflect natural research progression